

General Dwell Time Hidden Semi-Markov Models: Is it worth the trouble?

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Introduction

- Hidden Markov Models (HMMs) are widely used for modeling systems that evolve through an unobserved sequence of discrete states.
- A fundamental limitation of the standard HMM is that state durations are implicitly geometric.
- Hidden Semi-Markov Models (HSMMs) generalize HMMs by allowing explicit specification of state duration distributions.
- Each state is now capable of using any suitable distribution to represent the duration of stay in the state before transitioning to a new state.

Question: Too Much Flexibility?

Given the ability to choose from a variety of sojourn distributions for each state, is it worth going through the extra complexity to find good initialization parameters, develop software, and decipher meaning from the model? Or is there a single sojourn distribution flexible enough to work in all practical settings?

Quick Notation Overview

Let $C_t \in \{1, \dots, m\}$ denote the latent state process associated with transition probability matrix (t.p.m) Ω and X_t the observed process. An HSMM has the following parameters:

- T.p.m. Ω , $\omega_{i,j} = P(C_t = j | C_{t-1} = i, C_t \neq j)$.
- State dependent sojourn distributions $d_i(r)$, $r \in \mathbb{N}$.
- State dependent emission distributions $p_i(x) = P(X_t = x | S_t = i)$.

The associated HMM has the following parameters:

- T.p.m. Γ , $\gamma_{i,j} = P(S_t = j | S_{t-1} = i)$.
- State dependent emission distributions $p_{I_k}(x) = P(X_t = x | S_t \in I_k)$ (this notation will be handy soon).

How do we fit them? HSMM Representation as a HMM

Following the procedure developed in [2] we represent a HSMM as a HMM. Each HSMM state is expanded into a collection of HMM substates. Let

$$I_k = \left\{ n : \sum_{i=0}^{k-1} m_i < n \leq \sum_{i=0}^k m_i \right\}, \quad k = 1, \dots, m \quad (I_i = \{I_i^-, \dots, I_i^+\}),$$

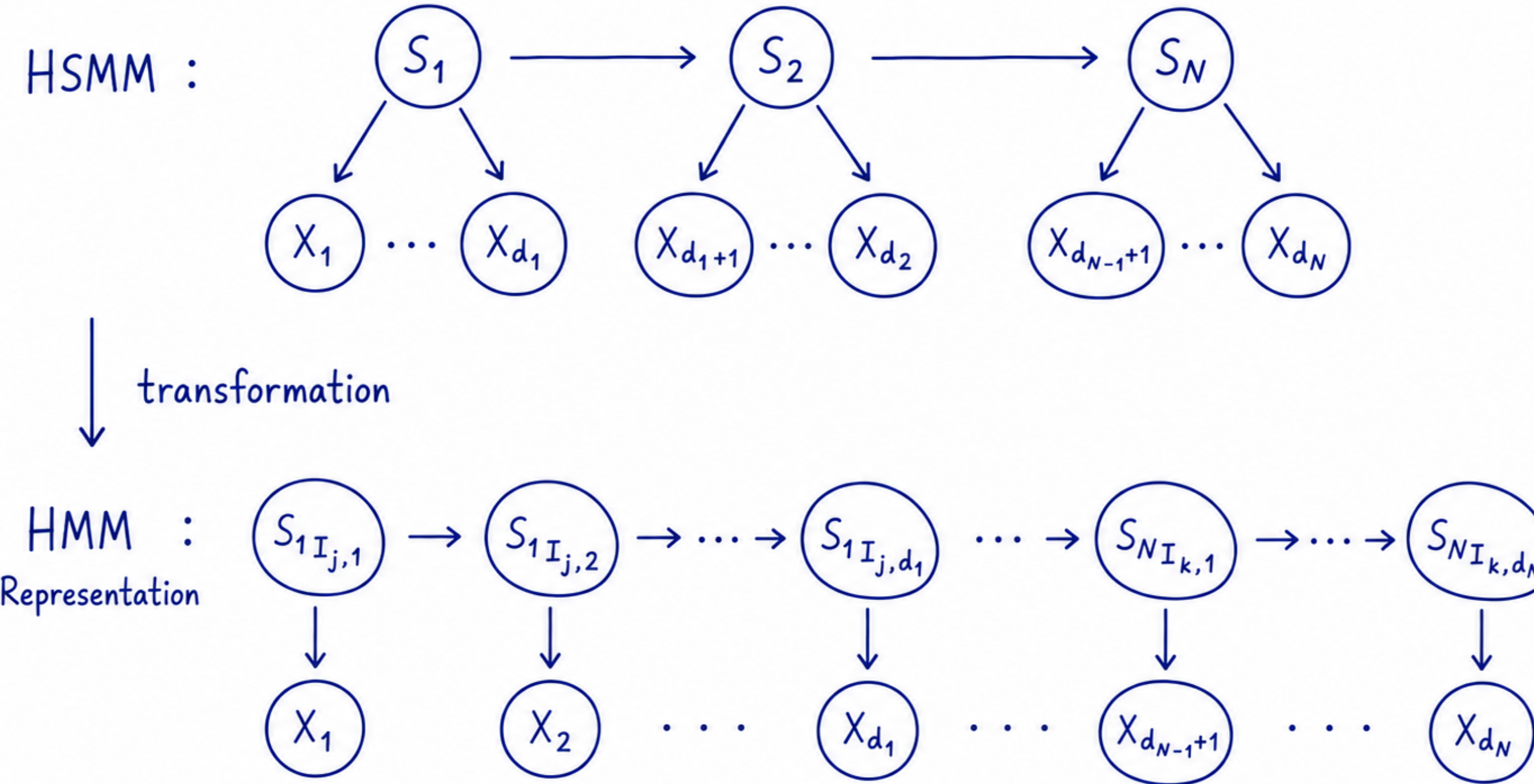
denote the aggregate corresponding to HSMM state i . Within each aggregate we embed the HSMM sojourns into blocks of a HMM t.p.m. The entries of each block are as follows:

$$c_i(r) = \frac{d_i(r)}{1 - D_i(r-1)}, \quad \omega_{ij} = P(C_{t+1} = j | C_t = i, C_{t+1} \neq j). \quad (1)$$

The entries of the HMM t.p.m. then become,

$$\mathbf{\Gamma}_{ii} = \begin{bmatrix} 0 & 1 - c_i(1) & \dots & 0 \\ 0 & 0 & 1 - c_i(2) & \dots & 0 \\ 0 & & \ddots & & \\ 0 & & & 1 - c_i(m_i - 1) & \\ 0 & & & 1 - c_i(m_i) & \end{bmatrix}, \quad \begin{bmatrix} c_i(1)\omega_{ij} & 0 & \dots & 0 \\ c_i(2)\omega_{ij} & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ c_i(m_i)\omega_{ij} & 0 & \dots & 0 \end{bmatrix}. \quad (2)$$

Using this construction we have a way to easily build and fit HSMMs with any choice of sojourn distribution without the overhead of creating custom code for each HSMM formulation.



Likelihood Evaluation Becomes Simple

Once expressed as a HMM, the likelihood may be written as

$$L_T = \delta P(x_1) \Gamma P(x_2) \Gamma \dots P(x_T) \mathbf{1}.$$

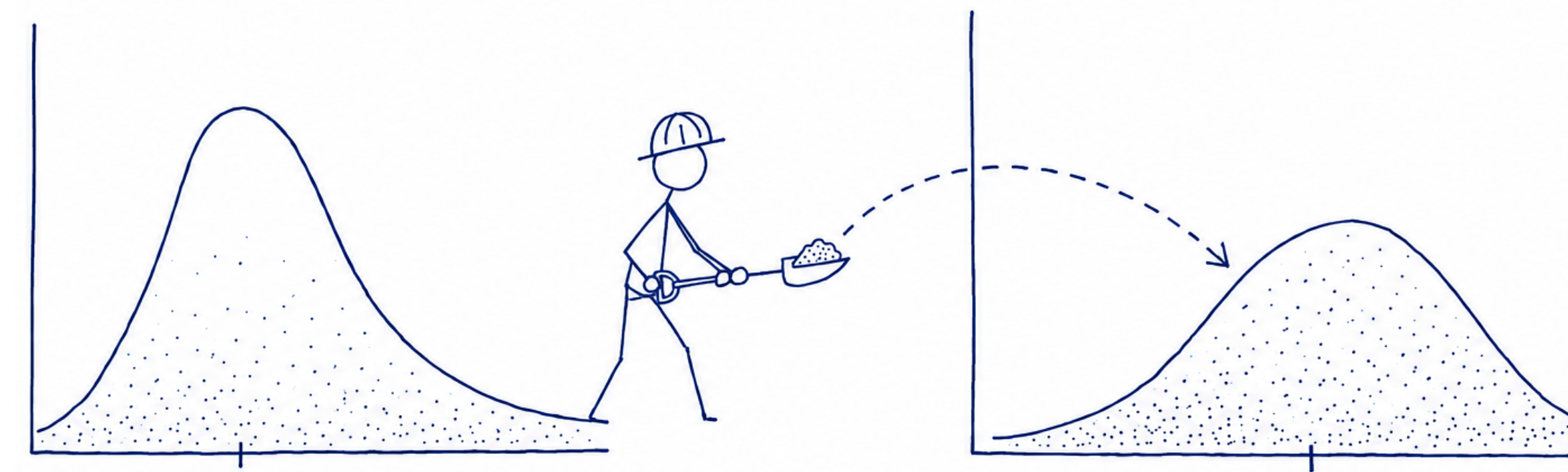
The forward recursion is: $\alpha_t = \alpha_{t-1} \Gamma P(x_t)$, with, $L_T = \alpha_T \mathbf{1}$.

HSMM $\rightarrow$ Expanded HMM $\rightarrow$ Forward Algorithm $\rightarrow$ Log-Likelihood

Once reformulated the HSMM likelihood can be numerically optimized with your favorite software!

How to Compare Sojourn Distribution?

Optimal transport provides a framework for comparing probability distributions by measuring the cost of transporting one distribution to another. The problem can be interpreted as moving one pile of sand into a desired shape while minimizing the total amount of work required.



Suppose μ and ν represent two distributions. A transport plan $\pi \in \Pi(f, g)$ specifies how we move the probability mass from μ to ν . If moving a unit of mass incurs a cost then the goal is to find the transport plan that minimizes the cost. The resulting minimum cost defines the Wasserstein distance.

$$W_2(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{X \times X} \|x - y\|^2 d\pi(x, y) \right)^{1/2},$$

$$\Pi(\mu, \nu) = \{ \pi : \pi(A \times X) = \mu(A), \pi(X \times B) = \nu(B) \}.$$

Intuition: the average amount of probability mass that must be moved by the distance over which it is transported.

In one dimension the Wasserstein-2 distance can be computed using quantile functions:

$$W_2^2(\mu, \nu) = \int_0^1 (F_\mu^{-1}(u) - F_\nu^{-1}(u))^2 du.$$

This works with any two distributions, continuous or discrete.

Intuition: Wasserstein distance is comparing the location of probability mass.

Simulation Study

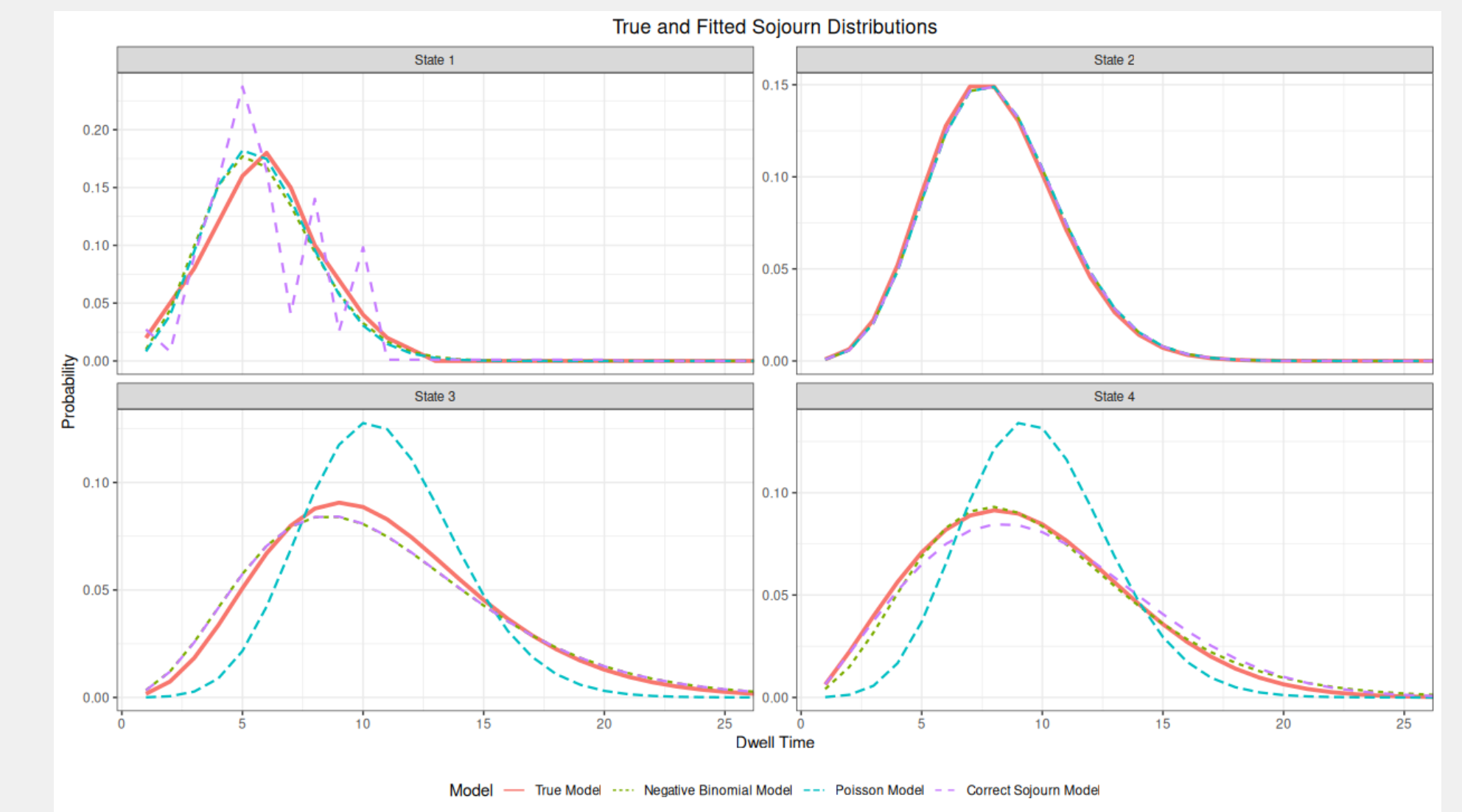
Data Generation: We generate some nasty data from a HSMM whose sojourn distributions are:

- State 1: Non-Parametric ($\Theta = (0.02, 0.05, 0.08, 0.12, 0.16, 0.18, 0.15, 0.10, 0.07, 0.04, 0.02, 0.01)$),
- State 2: Shifted Poisson ($\lambda = 7$, $shift = 1$),
- State 3: Shifted Negative Binomial ($p = 0.45$, $r = 8$, $shift = 1$),
- State 4: Weibul ($\lambda = 2.2$, $k = 10$),

emission distributions are Normal (μ, σ) and the t.p.m. is uniform(ish).

Experiment: Initialize three HSMMs: *True Model* where the known distributions are used, *Negative Binomial Model* where negative binomial sojourns are used for all states, and *Poisson Model* where Poisson sojourns are used for each state. Initial parameters are created through k-means clustering of observations and `fitdist()` [1] according to the schema developed in [3].

Results: Below is a plot of true versus fitted distributions, a table of corresponding W_2^2 distances from the fitted to true distributions, and Viterbi decoded sequences.



State	Negative Binomial	Poisson	Correct Sojourn
1	0.480	0.493	0.749
2	0.325	0.337	0.318
3	0.642	1.637	0.634
4	0.814	1.452	0.747
Viterbi Accuracy	0.996	0.995	0.996

Conclusion

It appears as though the Negative Binomial distribution preforms very well with the Poisson preforming the worst. This is further backed by the W_2^2 distances of fitted to true distributions. Most notably the NB distribution out preforms the true distributions for state 1 in terms of W_2^2 distance to the data generator distribution. Interestingly enough all 3 models have nearly perfect and identical Viterbi accuracy.

References

- [1] Marie Laure Delignette-Muller and Christophe Dutang. fitdistrplus: An R package for fitting distributions. *Journal of Statistical Software*, 64(4):1–34, 2015.
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- [3] Antonello Maruotti and Antonio Punzo. Initialization of hidden markov and semi-markov models: A critical evaluation of several strategies. *International Statistical Review*, 89(3):447–480, 2021.
- [4] Gabriel Peyré and Marco Cuturi. *Computational Optimal Transport*. Now Publishers, 2019.